higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

BUILDING AND STRUCTURAL SURVEYING N5

15 November 2019

This marking guideline consists of 4 pages.

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QUESTION 1

1.1 False
1.2 False
1.3 True
1.4 True
1.5 False

(5 × 2) **[10]**

QUESTION 2

2.1 Chaining is a common site name✓ given to all taping work.✓

2.2 It is horizontal direction✓ between any two points✓ defined in degrees, minutes and seconds.

2.3 Cadastral surveying is a branch of survey, related to the preparation of plans and✓ showing and defining legal property boundaries.✓

2.4 Parallax is when the cross hairs in the telescope and sighted object✓ are not seen in the same line✓ resulting in a blurred image of one or the other.

2.5 Booking is entering all survey measurements✓ in a tache book✓ and arranging them in such a manner that any surveyor may produce a map from them.

(5 × 2) **[10]**

QUESTION 3

3.1
- From the site boundaries
- measure out the proposed building
- increasing the area by ±1 m.
- Punch in two pegs (±2 m long pegs) 1 m away
- from each corner
- in line with the building line in all four corners.
- Because of the length of the pegs, a traveller of 1,5 m would be appropriate.
- The formation level plus/minus the benchmark
- plus the length of the traveller
- will give the staff reading on all the sight rails of the eight pegs. (10)

3.2
- Reading the staff upwards instead of downwards.
- Reading an inverted staff downwards instead of upwards. (2 × 2) (4)

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3.3 Each side = ✓25 m^2✓
= 5 m✓
= 5 000 mm✓
= 5 000 mm/200✓
= 25 mm✓ (6)

3.4
- The contours are closer together when the terrain has a steep slope. (4)
- The contours are further apart when the terrain has a gentle alope. (2 × 2) **[24]**

QUESTION 4

4.1
- Set the instrument on equal distances
- between two pegs A and B
- that are ± 600 m apart on a fairly level ground.
- Take staff readings at A and B and find the difference in height.
- Set the level closer to peg A and take staff readings,
- if the difference in height in both instances
- is the same, the instrument is right
- but if there is a difference then the difference is used to adjust the collimation line until it is horizontal.
- The adjustment is done by using the diaphragm adjusting screws until the staff reading is as calculated. (9)

4.2 **ADDENDUM - TABLE**

POINT	BACK SIGHT	INTER SIGHT	FORE SIGHT	RISE	FALL	REDUCED LEVEL	REMARKS
A	0,49					1020,600✓	
	0,27		3,29		2,800✓	1023,400✓	
	0,39		3,77		3,500✓	1026,900✓	
	3,72		3,59		3,200✓	1030,100	
		1,11		2,610✓		1027,490Š	
	3,56		0,82	0,290✓		1027,200✓	
	3,89		1,36	2,200✓		1025,000	BM 1025,00
	3,72		0,99	2,900✓		1027,900✓	
	3,69		1,02	2,700✓		103,600✓	
	3,86		1,31	2,380✓		1032,980	
	3,90		1,56	2,300✓		1035,280✓	
B			2,40	1,500✓		1036,780✓	

(20)
[29]

QUESTION 5

5.1 A–B = 89,288 m × Cos 2,506°✓
= 89,203 m✓

B–C = 100,020 m × Cos 3,675°✓
= 99,814 m✓

C–D = 67 m × Cos 1,420^0✓
= 66,979 m✓

A–D = 89,203 m✓ + 99,814 m✓ + 66,979 m✓
= 255,996 m✓ (10)

5.2 ∆h A–B = 127 m – 126 m
= 1 m✓

∆L A–B = 36 m – 0
= 36 m✓
Tan ø = 1/36
= 0,027✓
Therefore the slope is 1:36✓

∆h B-C = 126 m – 126 m
= 0✓
∆L B-C = 53 m – 36 m
= 17 m✓
Tan ø = 0/17 m
= 0✓
The slope is zero and therefore the terrain is horizontal✓✓

∆h C-D = 125 m 124 m
= 1 m✓
∆L C-D = 100 m – 53 m
= 47 m✓
Tan ø = 1/47 m
= 0,021✓
Therefore the slope is 1:47✓

∆h D-E = 124 m – 120 m
= 4 m✓
∆L D-E = 297,80 m – 100 m
= 197,80 m✓
Tan ø = 4/197,80
= 0,02022✓
Therefore the slope is 1:49✓ (17)
[27]

TOTAL: 100

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